

SIMATS ENGINEERING

ASSIGNMENT-2

COURSE: MACHINE LEARNING (ITA 0603)

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Question: Represent a neural network architecture (input, hidden, output layers) suitable for predicting crop yield from soil moisture, temperature, and rainfall data. Justify the number of neurons chosen. (SDG 2 – Zero Hunger)

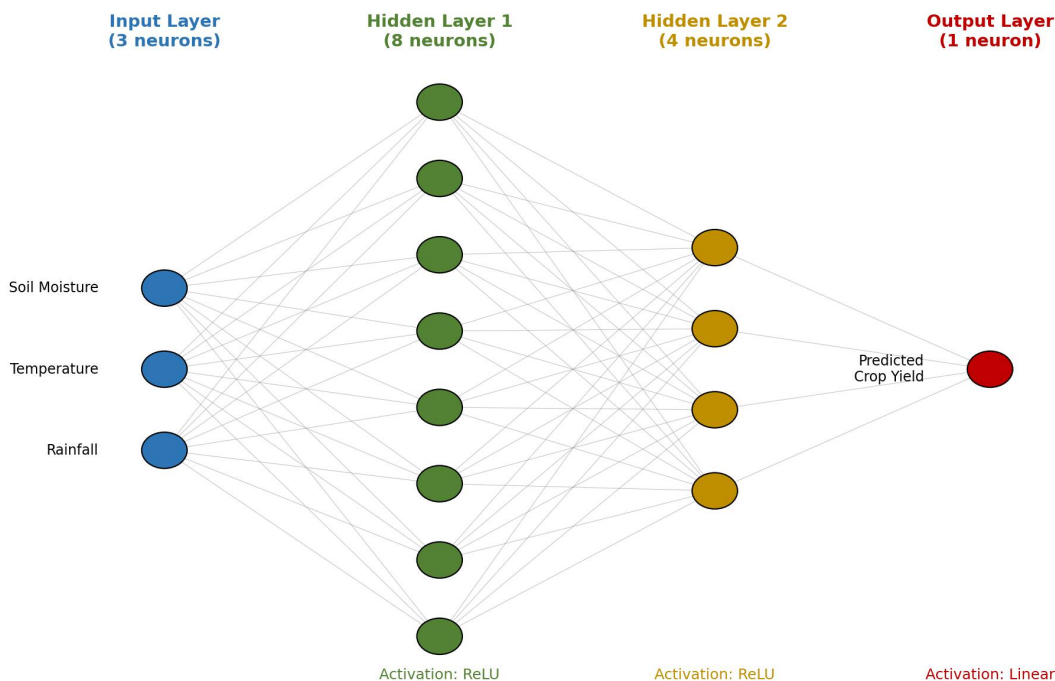
Answer:

1.Problem Framing

Crop yield (e.g., tons per hectare) is a continuous numeric quantity influenced by soil moisture, temperature, and rainfall. This is therefore modeled as a regression problem: given the three input readings, the network outputs a single predicted yield value. A fully-connected (dense) feed-forward neural network — a Multilayer Perceptron (MLP) — is well suited, since the relationship between weather/soil conditions and yield is non-linear (e.g., yield rises with rainfall only up to a point, after which excess rainfall can reduce it).

2.Proposed Neural Network Architecture

Neural Network Architecture for Crop Yield Prediction



Layer-by-Layer Summary:

Layer	Neurons	Activation	Purpose
Input Layer	3	—	Receives the three raw features: soil moisture, temperature, rainfall.
Hidden Layer 1	8	ReLU	Learns broader non-linear combinations/interactions of the three inputs.
Hidden Layer 2	4	ReLU	Condenses the learned features into a more compact, higher-level representation.
Output Layer	1	Linear	Produces the single continuous predicted crop yield value.

3. Justification of Neuron Counts

Input Layer — 3 neurons:

The number of input neurons is fixed by the number of available features. Since the problem specifies exactly three measured variables — soil moisture, temperature, and rainfall — the input layer must have exactly 3 neurons, one per feature, with no more and no less.

Hidden Layer 1 — 8 neurons:

A common rule of thumb is to set the first hidden layer's width to roughly 2–3 times the number of input features, giving the network enough representational capacity to combine the 3 raw inputs into a richer set of intermediate features (e.g., 'moisture × temperature interaction', 'rainfall adequacy index') without introducing excessive parameters. With only 3 inputs, 8 neurons (~2.7×) is large enough to capture non-linear interactions between soil moisture, temperature, and rainfall, yet small enough to avoid overfitting on what is typically a modest-sized agricultural dataset.

Hidden Layer 2 — 4 neurons:

The second hidden layer follows a funnel/tapering design, progressively reducing the number of neurons (8 → 4) as the network moves toward the output. This forces the model to compress the broader set of features learned in Hidden Layer 1 into a smaller number of high-level, more abstract features that are most relevant for predicting yield, which improves generalization and reduces the risk of overfitting compared to keeping the width constant or increasing it.

Output Layer — 1 neuron:

Since the task is to predict a single continuous value — the crop yield — only one output neuron is required, using a linear (identity) activation function so the network can output any real-valued yield estimate rather than being restricted to a bounded range (as sigmoid/softmax would impose).

Activation Functions:

- ReLU (Rectified Linear Unit) is used in both hidden layers because it is computationally efficient, helps avoid the vanishing-gradient problem seen with sigmoid/tanh, and works well for capturing the non-linear thresholds often present in agronomic relationships (e.g., yield loss beyond a rainfall saturation point).
- A linear activation is used at the output layer since crop yield is an unbounded, continuous regression target.

General Sizing Principle Applied:

Overall, the architecture follows the widely used heuristic of keeping hidden-layer width between the size of the input layer and a modest multiple of it (avoiding layers so large that the model overfits a

small tabular dataset), combined with a tapering structure across hidden layers so the network progressively distills the three raw environmental readings into an increasingly refined, yield-relevant representation before the final prediction.

4. SDG 2 – Zero Hunger Relevance

This model directly supports UN Sustainable Development Goal 2 (Zero Hunger), which calls for ending hunger, achieving food security, and promoting sustainable agriculture.

- Early Yield Forecasting – predicting yield ahead of harvest helps farmers and governments anticipate potential shortfalls and plan food distribution or imports accordingly.
- Precision Agriculture – by learning how soil moisture, temperature, and rainfall jointly affect yield, the model can guide targeted irrigation and resource allocation, improving productivity without wasting water or inputs.
- Climate Resilience – understanding the non-linear effect of rainfall/temperature extremes on yield supports early warning systems for climate-related crop stress, helping safeguard food security in vulnerable regions.
- Sustainable Resource Use – optimized predictions reduce over-irrigation and over-fertilization, supporting environmentally sustainable farming practices alongside higher, more reliable yields.